Module 7

I²C — Protocol + RTL + Basic Testbench

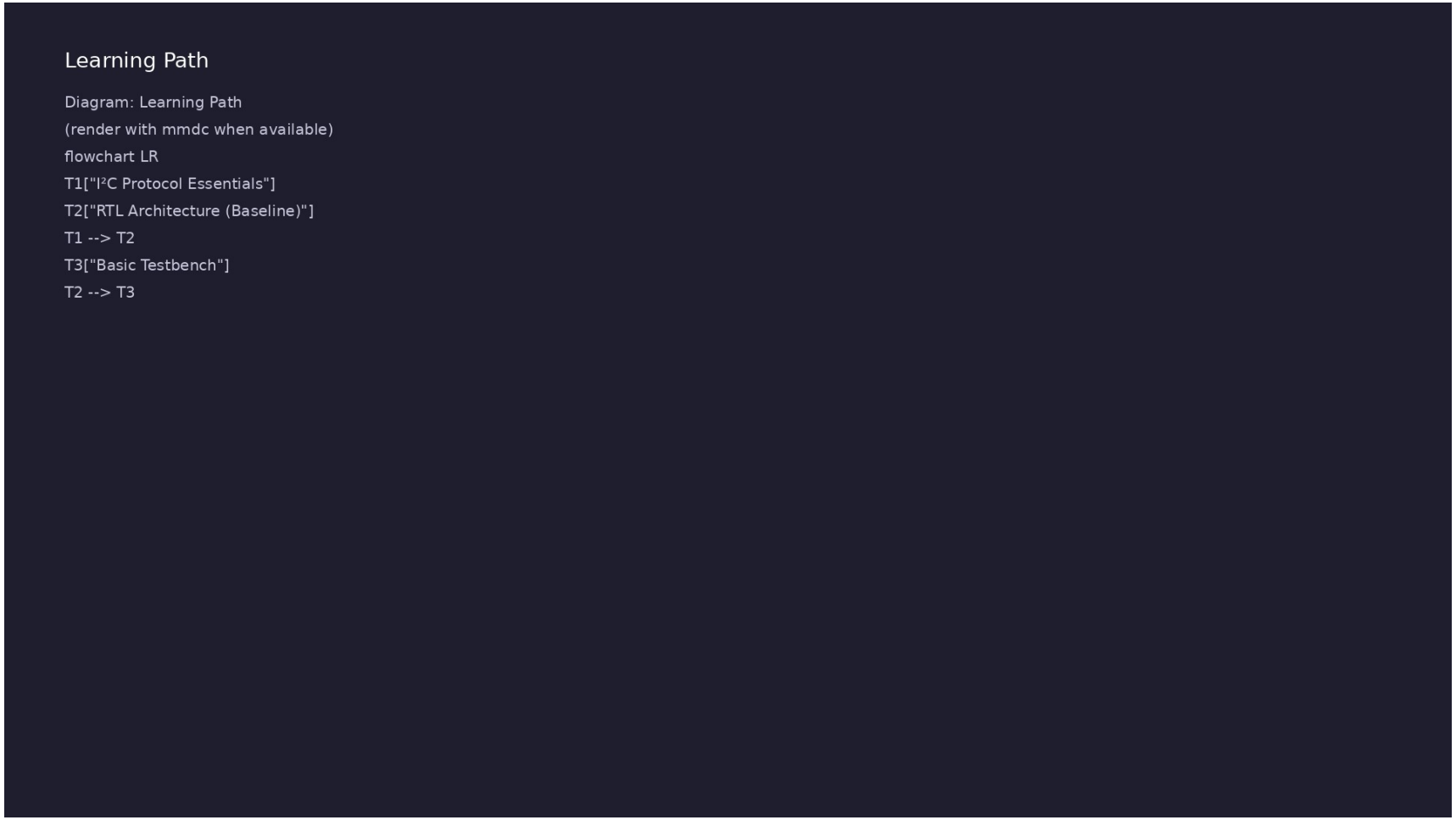
Learning objectives

- How START/STOP are detected/produced (SDA transitions while SCL high).
- How the address + R/W bit and data bytes are serialized on SDA.
- How to build a basic “bus monitor” that reconstructs bytes by sampling SDA on SCL rising edges.

Prerequisites

- Modules 1–6.
- Verilator, Make, C++ compiler. No UVM required for this baseline.

Learning path



Understand the I²C protocol (start/stop, addressing, ACK/NACK), translate it to RTL (simple master + timing), and verify with a **basic** (…

Overview

- Module 7 is the I²C **protocol + RTL + basic testbench** module (UVM comes in Module 8).

Syllabus topics

1. I²C Protocol Essentials

- Bus idle: SCL=1, SDA=1.
- START: SDA goes 1→0 while SCL is 1.
- STOP: SDA goes 0→1 while SCL is 1.
- Address phase: 7-bit address + R/W bit (0=write, 1=read) sent MSB first.
- Data phase: 8-bit data bytes sent MSB first.
- ACK/NACK (concept): receiver pulls SDA low for ACK on the 9th clock; high means NACK.

2. RTL Architecture (Baseline)

- clk_div: Divides `clk` to produce `clk_div_tick`
  used to toggle/advance SCL timing.
- i2c_master: State machine that generates:
- START condition
- address+W bits
- one data byte
- STOP condition

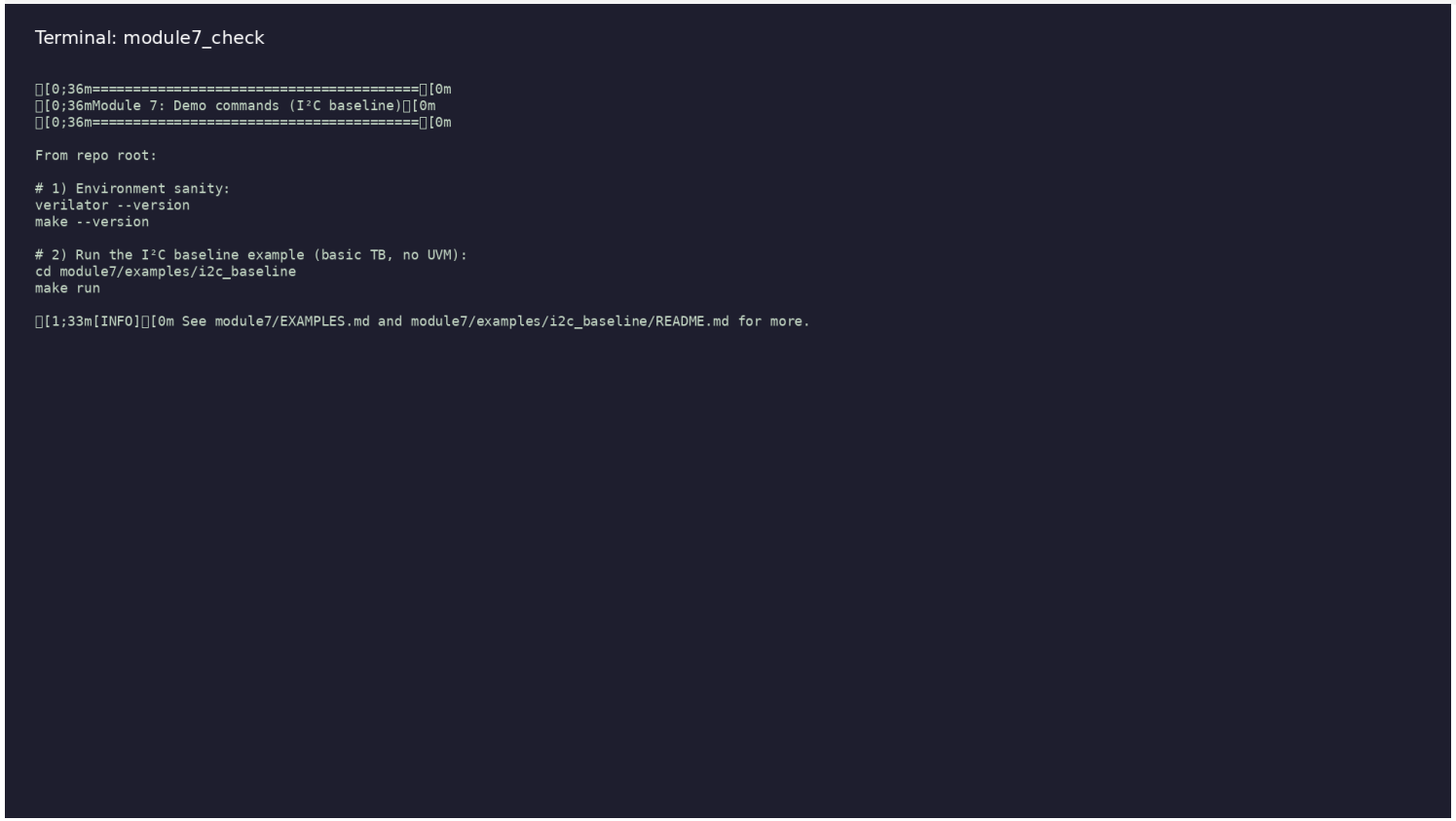
3. Basic Testbench

- Stimulus: Pulses `start`, sets `addr` and `data_in`, waits for `done`.
- Monitor/self-check: Detects START, then samples SDA on each rising edge of SCL to reconstruct:
- 8 bits of address+W
- 8 bits of data

Hands-on examples

Module 7 self-check

```
$ ./scripts/module7.sh --check
```



Example 1: I²C baseline

- Drives a single write transaction (START → address+W → data → STOP)
- Includes a simple bus monitor that samples SDA on SCL rising edges to reconstruct bytes
- No UVM

Demo: I²C baseline

```
$ ./scripts/module7.sh --run
```

Practice & assessment

Exercises

- Run i2c_baseline
- Trace one transfer
- Bus monitor
- Optional: Waveforms

Assessment checklist

- Can describe I²C START, STOP, address+R/W, and data phase (and ACK/NACK concept).
- Can explain the role of i2c_master and clk_div; know the baseline uses push-pull (simplified).
- Can run i2c_baseline and interpret the test result.
- Ready for Module 8: I²C UVM+SV verification.

Summary & next steps

- Understand the I²C protocol (start/stop, addressing, ACK/NACK), translate it to RTL (simple master…
- Complete module7/CHECKLIST.md
- Run ./scripts/module7.sh --check
- Next: I²C UVM+SV